18/11/20

$q \,,\, \vec{v} \,,\, \vec{B} \,,\, \vec{E}$

$\theta = \vec{v} \measuredangle \vec{B}$

$$\vec{F}_m = q(\underbrace{\vec{v} \times \vec{B}}_{\text{LORENTZ}} + \vec{E})$$

- $\underline{\theta = \pi/2}$ $(\vec{v} \perp \vec{B})$

$$F_m = qvB = m a_n = m\frac{v^2}{r}$$

$$\Rightarrow \boxed{r = \frac{mv}{qB}} \qquad \left(B = \frac{mv}{qr} \right)$$

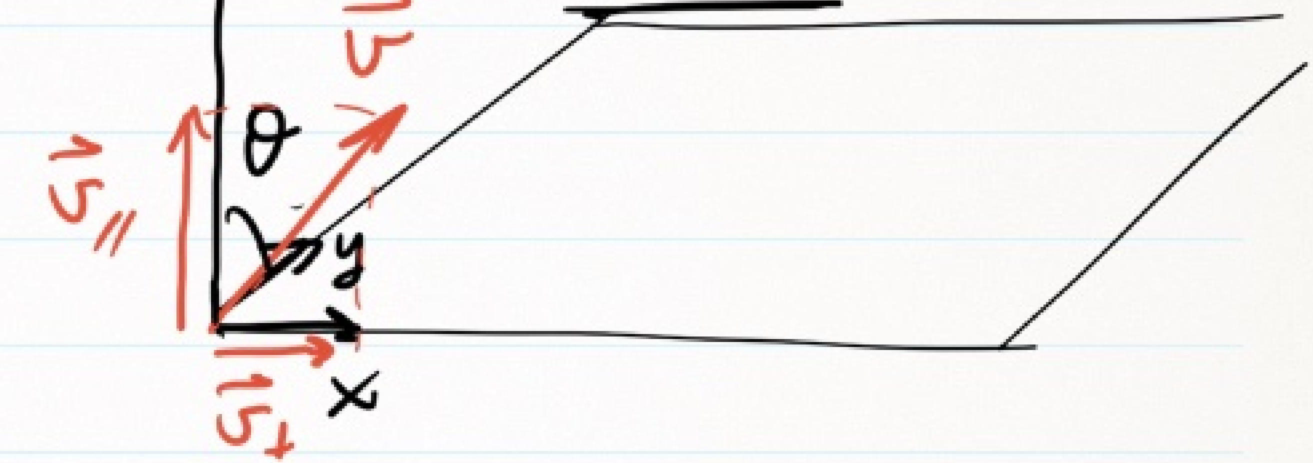
(i) @ $v = \text{cost} \wedge B = \text{cost}$ $\quad r \propto m/q$

(ii) $r \propto v$, $\omega = \frac{v}{r} = \frac{qB}{m}$ @ $B = \text{cost}$ $\underline{\omega} \propto q/m$

$$\boxed{T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}} \qquad \boxed{\vec{\omega} = -\frac{q}{m}\vec{B}}$$

• $\underline{\theta \neq \pi/2}$

$\underline{\vec{B} = B\hat{z}}$



$$F_m = q \underbrace{v\sin\theta}_{|\vec{v}_{\perp}|} B$$

$$m a_n = m\frac{(v\sin\theta)^2}{r} \Rightarrow \boxed{r = \frac{m v \sin\theta}{qB}}$$

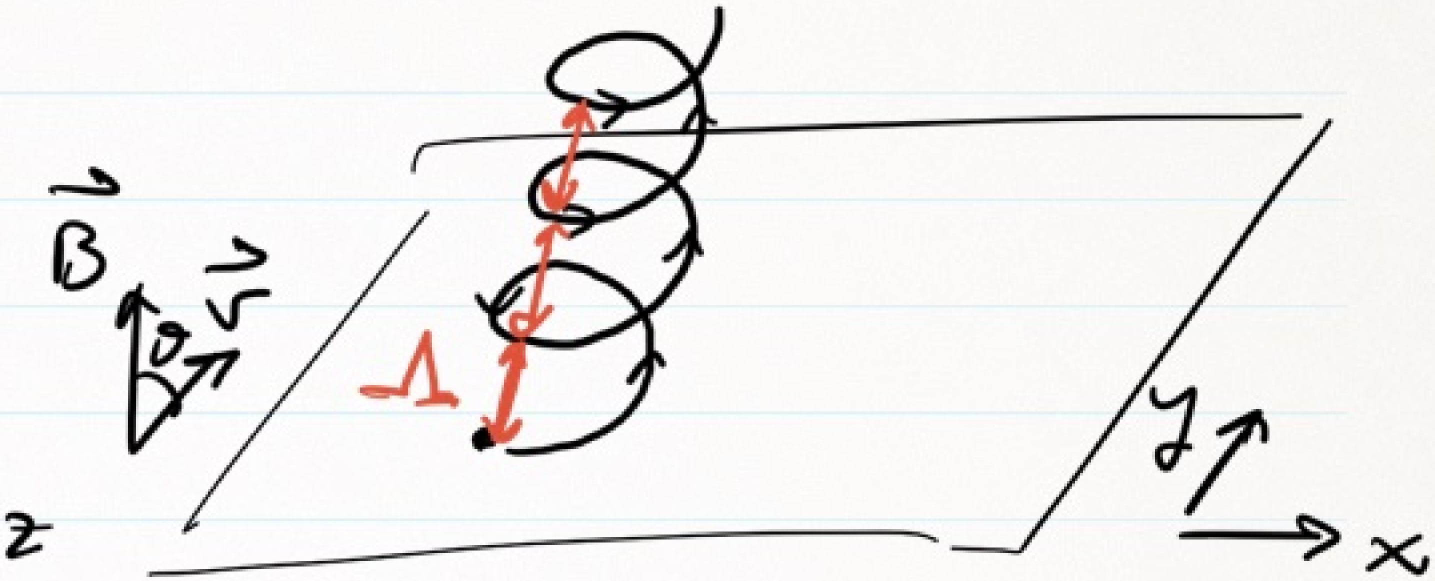
$$\omega = \frac{v\sin\theta}{r} = \frac{qB}{m}$$

$$|\vec{v}_{\parallel}| = v\cos\theta$$

m.c.u.$|_{xy}$ +

m.r.u.$|_{z}$ =

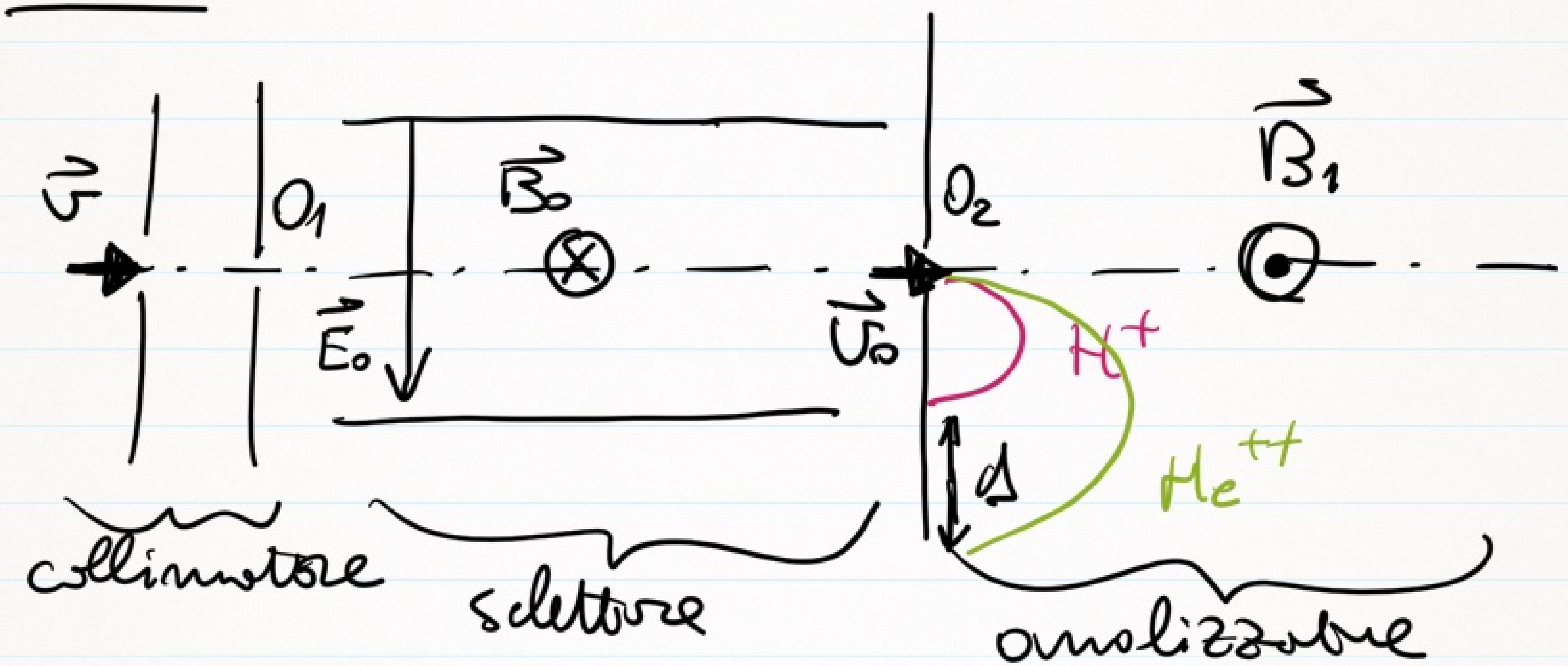
moto elicoidale $|_{xyz}$



$$\Lambda = v_{//} \cdot T = \frac{v \cos\theta \, 2\pi m}{qB}$$ Passo dell'elica

$$\begin{cases} F_x = m \dfrac{d^2x}{dt^2} = q \dfrac{dy}{dt} B \\ F_y = m \dfrac{d^2y}{dt^2} = -q \dfrac{dx}{dt} B \\ F_z = m \dfrac{d^2z}{dt^2} = 0 \end{cases}$$

Ex. 1 SPETTROMETRO DI MASSA



(a) $\cancel{q} E_0 = \cancel{q} v_0 B_0 \Rightarrow v_0 = \dfrac{E_0}{B_0} = 10^5 \dfrac{m}{s}$

(b) $d = 2r = 2 m v_0 / q B_1$

$$d_{H^+} = \frac{2 \cdot 1.67\,10^{-27}\,10^5}{1.6\,10^{-19}\,0.5} = 4.18\,mm$$

$$d_{He^{++}} = 2\,d_{H^+} = 8.36\,mm$$

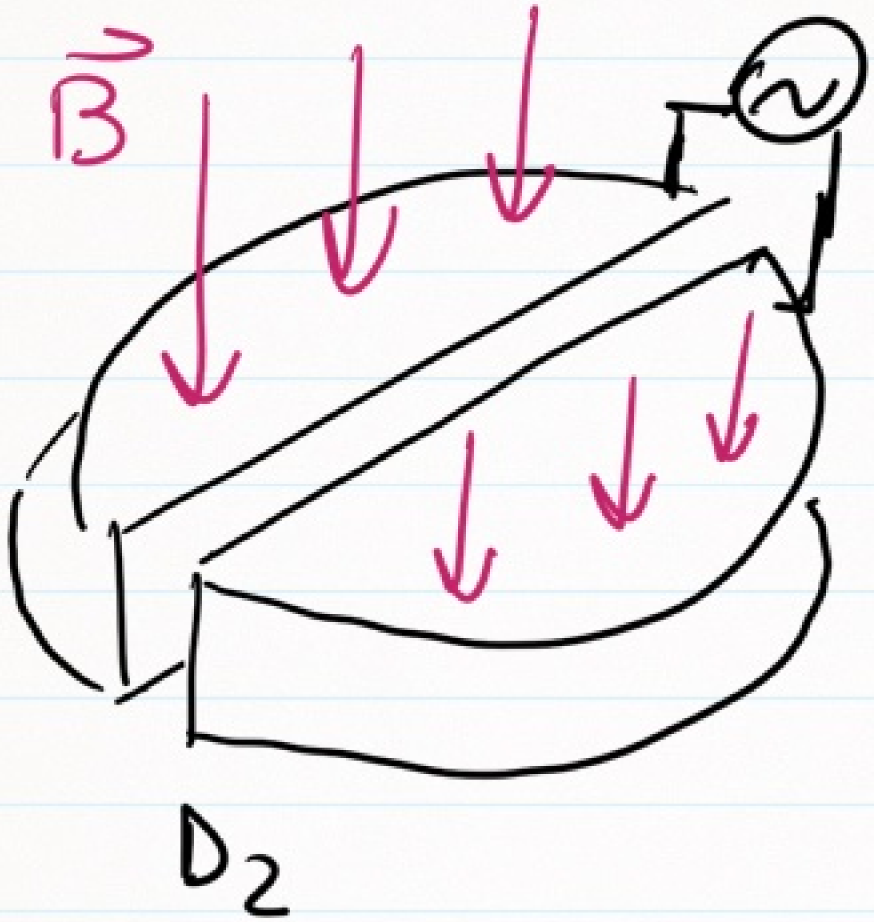
$$\Delta = d_{He^{++}} - d_{H^+} = 4.18\,mm$$

ΔV fisso

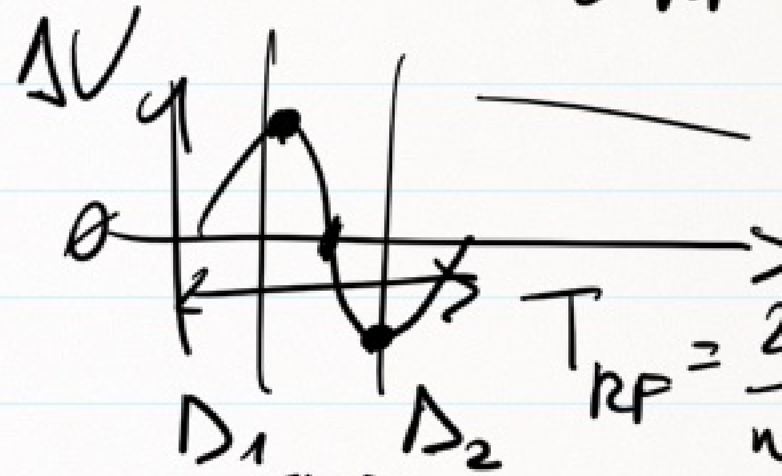
$$q\Delta V = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{\frac{2q\Delta V}{m}}$$

<u>acceleratore lineare</u>

Ex. 2



$$\Delta V = \Delta V_0 * \sin(\omega_{RF} t)$$



$$T_{RF} = \frac{2\pi}{\omega_{RF}}$$

$$\boxed{\omega_{RF} = \frac{qB}{m}}$$

q

@ t = 0

$$v_1 = \sqrt{\frac{2q\Delta V}{m}} \quad , \quad r_1 = \frac{m v_1}{qB}$$

@ t = t_1 $= \frac{T}{2} = \frac{\pi m}{qB}$

$$\underbrace{\frac{1}{2} m v_1^2}_{\underline{q\Delta V}} + \underline{q\Delta V} = \underbrace{\frac{1}{2} m v_2^2 = 2\,q\Delta V}$$

in D_2 @ $v_2 = \sqrt{\dfrac{4q\Delta V}{m}} = \sqrt{2}\cdot v_1$

$$r_2 = \sqrt{2}\, r_1$$

@ $\underline{t = t_2} = t_1 + \dfrac{T}{2} = 2t_1$

⋮ ⋮

n-esimo passaggio (t_n):

$$\frac{1}{2} m v_n^2 = n \cdot q \Delta V$$

$$v_n = \sqrt{n} \cdot v_1$$

$$r_n = \sqrt{n} \cdot r_1 = \sqrt{\frac{2m\Delta V}{qB} \cdot n}$$

$$v_{max} \longrightarrow r = R$$

$$v_{max} = \frac{qBR}{m}$$

$$E_{K,max} = \frac{1}{2} m v_{max}^2 = \frac{(qBR)^2}{2m}$$

$$E_{K,max} = \frac{(2 \cdot 1.6 \cdot 10^{-19} \cdot 1.3 \cdot 0.5)^2}{2\, m_\alpha} = 3.26 \cdot 10^{-12}\, J$$

$$m_\alpha = 2m_{+e} + 2m_{neutroni} = 6.68 \cdot 10^{-27}\, kg$$

$$\left[1eV = 1.6 \cdot 10^{-19}\, J \;,\; E_{K,max} = 20.26\, MeV \right]$$

$$v_{max} = \frac{2 \cdot 1.6 \cdot 10^{-19}\; 1.3\;\, 0.5}{6.68 \cdot 10^{-27}} \simeq 3.11 \cdot 10^{7}\, \frac{m}{s}$$

$$v_{max} = 0.1 \cdot c$$

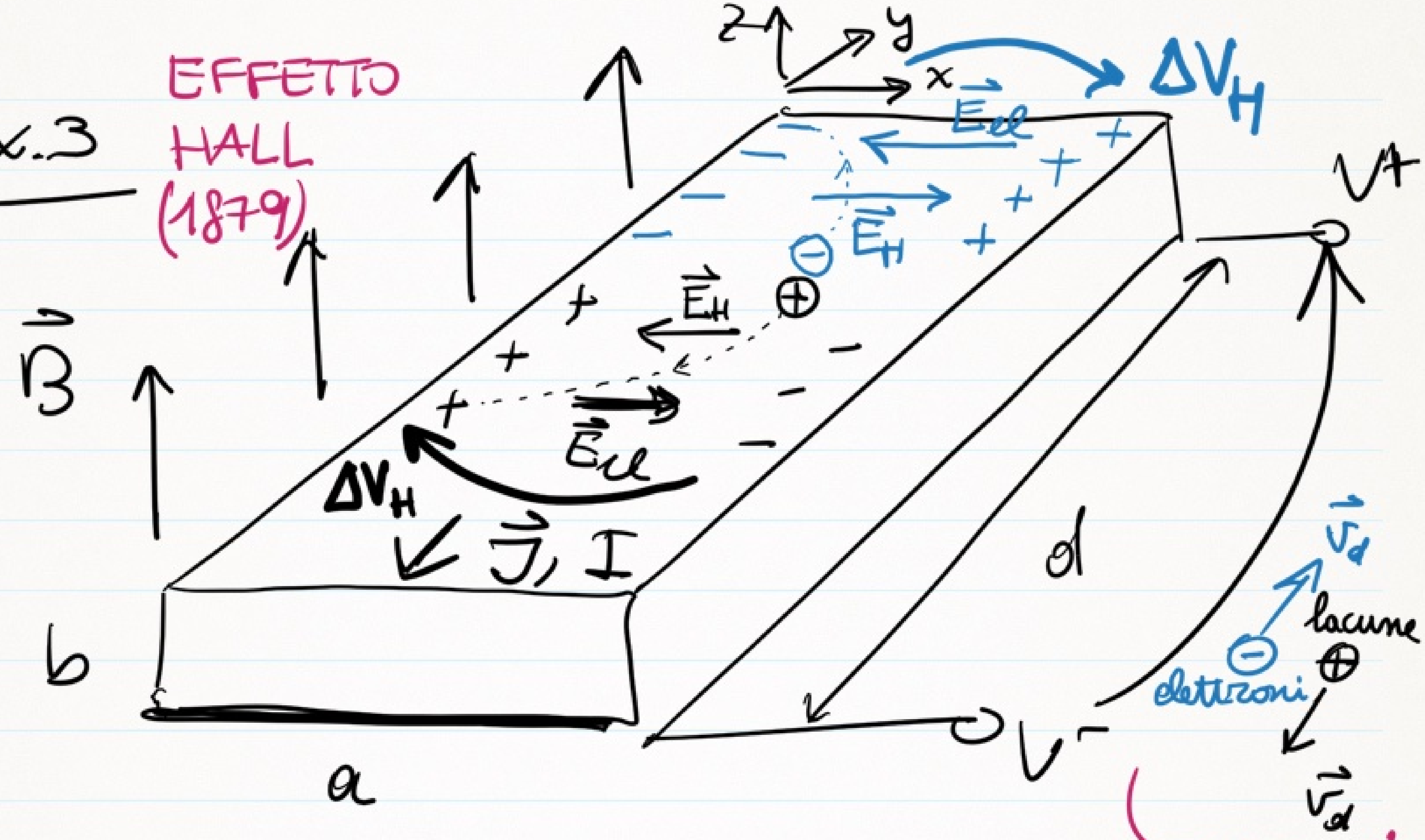
$$N_{giri} = \frac{E_{K,max}}{2 q \Delta V} = \frac{q (BR)^2}{4 m \Delta V}$$

$$= \frac{3.24 \; 10^{-12}}{2 \cdot 2 \cdot 1.6 \cdot 10^{-19} \; 10^4} \simeq 506 \simeq 500 \text{ giri}$$

$$T_{max} = N_{giri} \cdot T \simeq 0.51 \; \mu s$$

Ex. 3

EFFETTO HALL (1879)



$$\vec{J} = -\frac{I}{ab}\,\hat{y} = -\,n\,q\,v_d\,\hat{y}$$

$$\vec{F}_m = q\vec{v}_d \times \vec{B}$$

indipendentemente dal loro segno, i portatori si accumulano sullo stesso lato, MA posso distinguerli dal SEGNO di ΔV_H

CAMPO DI HALL : $\vec{E}_H = \dfrac{\vec{F}_m}{q} = \vec{v}_d \times \vec{B} =$

$$= \frac{\vec{J}}{nq} \times \vec{B}$$

$\vec{E}_H + \vec{E}_{el} = \vec{0}$ @ equilibrio

$|F_m| = |F_{el}|$, $|E_H| = |E_{el}| = \dfrac{J}{ne}B$

TENSIONE DI HALL : $\underline{\Delta V_H} = E_H \cdot a$

$$= \frac{JB}{nq} \cdot a$$

$$\Delta V_H = \underbrace{\frac{I}{\cancel{a}b} \cdot \frac{B}{nq}}\,\cancel{a} = \frac{(V^+ - V^-)}{\rho \frac{d}{a\cancel{b}}} \cdot \frac{B}{nq\cancel{b}} =$$

$$= \frac{(V^+ - V^-)}{\rho d} \cdot \frac{Ba}{nq} \leftarrow$$

$$\Delta V_H = \frac{1}{nq} \cdot \frac{IB}{b}$$

Resistenza di Hall

$$\boxed{R_H = \frac{1}{nq}}$$

(i) SEGNO DEI PORTATORI $R_H > 0$ lacune

$R_H < 0$ elettroni

$R_H \sim 10^{-11} \frac{m^3}{C}$ Cu, Ag, Au, Al

(ii) DENSITÀ VOLUMICA n PORTATORI

(iii) SONDA DI $\vec{B}$ (SONDA DI HALL)

$$\frac{\Delta V_H}{B} = \underbrace{\frac{(V^+ - V^-)a}{e\, d\, n\, q}}$$

$$\left[\frac{\Delta V_H}{B}\right] = \frac{V}{T}$$

(a) $\uparrow \Delta V_H = \dfrac{I\, B}{\downarrow b\, n\, q}$

$n_{Cu} = 8.49 \cdot 10^{28}\, m^{-3}$

$[\delta, MM, N_A] \leftarrow$

$n_{Cu} = N_A \cdot \dfrac{\delta}{MM}$

$$= \frac{5 \cdot 1.5}{0.1 \cdot 10^{-2}\; 8.49\, 10^{28} \cdot 1.6\, 10^{-19}}$$

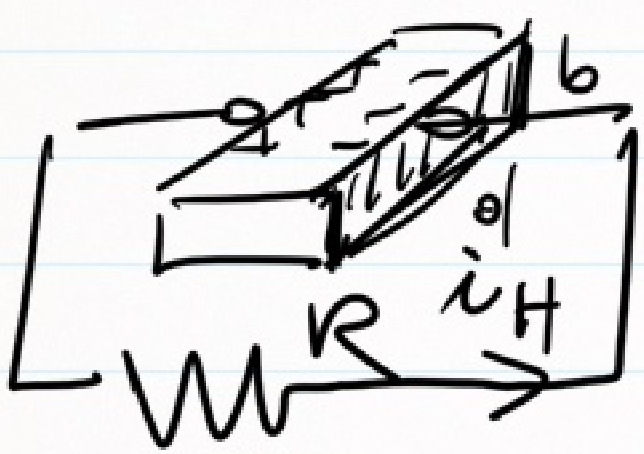
$\simeq \underline{0.55\ \mu V}$

$$\Delta V_{ohm} = R_{ohm} \cdot I = \rho \frac{\Delta y}{ab} \cdot I = \Delta V_H$$

$$\underline{\Delta y} = \frac{\Delta V_H}{\rho_{Cu} I} \cdot ab = \frac{0.55 \cdot 10^{-6}}{1.7 \cdot 10^{-8}} \, 1.5 \cdot 10^{-2} \, 0.1 \cdot 10^{-2}$$

$$= 10^{-4}\ \mathrm{m} = \underline{100\ \mu\mathrm{m}} \quad (!!)$$

(b)



$$i_H = \frac{\Delta V_H}{R + R}$$

$$R = \rho \cdot \frac{a}{bd} = 1.7 \cdot 10^{-8} \cdot \frac{1.5\ 10^{-2}}{0.5\ 10^{-2} \cdot 2 \cdot 10^{-2}} =$$

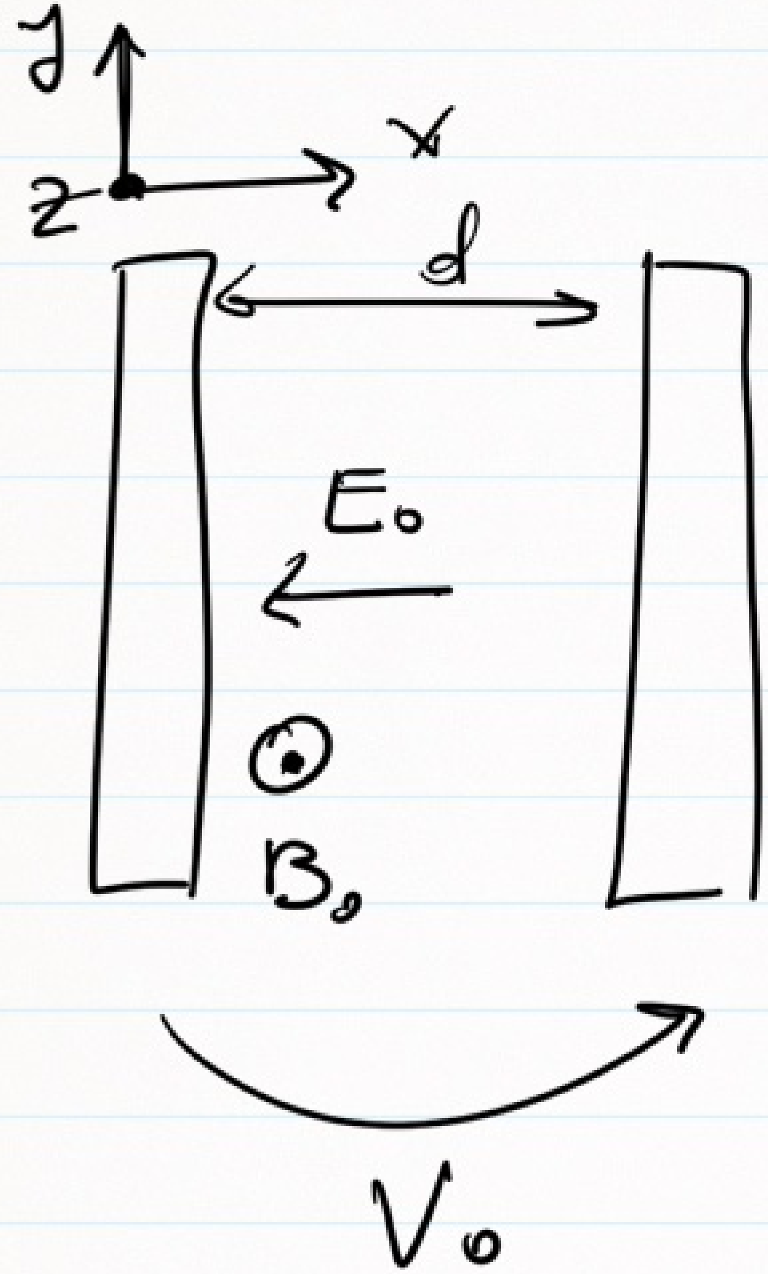
$$= 1.28 \cdot 10^{-5}\ \Omega$$

$$i_H = \frac{0.55\ 10^{-6}}{(0.128+1)\ 10^{-9}} = \underline{4.4\ mA}$$

$$\left[\frac{i_H}{B} = \frac{1}{r+R} \cdot \frac{I}{bnq} = 3.26\ \frac{mA}{T}\right]$$

sonda di campo B

Ex. 6



$$\vec{F} = q\left(\vec{E}_0 + \vec{v} \times \vec{B}\right)$$

$F_x = \dots$

$F_y = \dots$

$F_z = \dots$

$v_{x,y}(t)$

$x, y\,(t)$ (a)

(b) $$B_{min} = \sqrt{\frac{2 V_0 m_e}{d^2 e}}$$